



SIMATS ENGINEERING

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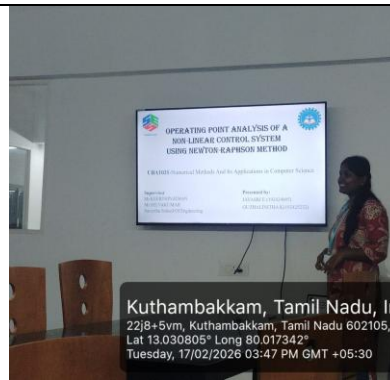
Slot: B

Course Name: Numerical Methods And Its Application In Computer Science

Course Faculty: Mr.KEERTHINATHAN And Mr.SELVAKUMAR

Project Title: OPERATING POINT ANALYSIS OF A NON-LINEAR CONTROL SYSTEM USING newton-Raphson method

Module Photographs:



Mathematical Model

Nonlinear System Model (State Space)

$$\dot{x} = f(x, u)$$

Where

$x \in \mathbb{R}^n \rightarrow$ State vector
 $u \rightarrow$ Input

Operating (Equilibrium) Point Condition

At operating point:

$$\dot{x} = 0$$

$$f(x_0, u_0) = 0$$

This gives a system of nonlinear algebraic equations.

Newton-Raphson Method (Multivariable Form)

$$x^{(k+1)} = x^{(k)} - J^{-1}(x^{(k)})f(x^{(k)})$$

Where:

$$F(x) = f(x, u)$$

$$J(x) = \frac{\partial F}{\partial x} \text{ (Jacobian Matrix)}$$

Final Mathematical Model

$$x_{k+1} = x_k - J^{-1}(x_k) f(x_k, u)$$

Project Description: Module 1- Data Interface and Data Acquisition

This module focuses on collecting and preparing the mathematical model required to determine the operating (equilibrium) point of a non-linear system using the Newton–Raphson method. The main purpose of this module is to acquire the system’s non-linear equations and parameters, generate suitable initial guesses, and iteratively compute the steady-state operating point for further analysis.

In this module, the system dynamics are represented in the form of non-linear algebraic equations derived from the underlying control or physical model. Since the project is implemented completely in software, the system variables and parameters are provided as numerical inputs or simulated data rather than measurements from physical hardware. The input equations and parameters are validated and pre-processed to remove inconsistent values, undefined variables, or incomplete model definitions. This step ensures that only well-formed and numerically stable inputs are passed to the Newton–Raphson iteration process.

The computed operating point is then forwarded to the subsequent analysis stages such as linearization, stability assessment, and controller design. The module also maintains basic logging of iteration counts, error norms, and convergence status to support monitoring, debugging, and verification of the numerical solution process.

Outcomes:

Overall, this module plays a fundamental role in enabling accurate operating point analysis of a non-linear system. By providing a reliable and converged steady-state solution using the Newton–Raphson method, this module establishes a strong foundation for linearization, performance evaluation, and stable control system design.

Student Signature

Guide Signature